IIT Bombay Nanofabrication Facility

Tool Name: Raith 150
Location: Nanolitho lab

Available Recipes

April 2025

Sr. No.	Recipe	Developed Recipe	Open Access/ Restricted	Contact for Recipe
1	PMMA-PMMA-bilayer metal lift-off detail recipe	IITBNF Recipe	Open Access	System Owner/Operator
2	PMMA 950 A4 single layer	IITBNF Recipe	Open Access	System Owner/Operator
3	PMMA 950 A4 -EL9 bilayer	IITBNF Recipe	Open Access	System Owner/Operator
4	PMMA 950 A2 -EL9 bilayer	IITBNF Recipe	Open Access	System Owner/Operator
5	HSQ- 250 C	IITBNF Recipe	Open Access	System Owner/Operator
6	HSQ- 90 C	IITBNF Recipe	Open Access	System Owner/Operator
7	SU8-2000.5	IITBNF Recipe	Open Access	System Owner/Operator
8	ZEP 520 A	IITBNF Recipe	Open Access	System Owner/Operator

Detail Recipes:

1. PMMA 495K A4 and PMMA 950K A4 bilayer with metal lift-off recipe

EBL process-

Sample- RCA cleaned 2 by 2 cm2 Si piece

1. Clean with Acetone (Sonicate) 1 min followed by IPA rinse.

2. Dehydration bake at 180 deg C for 5 min

3. Spin PMMA 495K A4

Step1- 5 sec, Speed-1000 rpm, Acceleration 500; Step2- 59 sec, Speed-4000 rpm , Acceleration: 2000

Pre-Bake at 180 deg C for 2mins.

4. Spin PMMA 950K A4

Step1- 5 sec, Speed-1000 rpm, Acceleration 500; Step2- 59 sec, Speed-4000 rpm , Acceleration: 2000, Pre-Bake at 180 deg C for 2mins.

5. Exposure- Voltage- 20 KV; Aperture -10 um; Write field -100 um; Steps-0.006; Area Dose- 160-180 uC/cm2 (please consult with process team for updated dose, as optimal dose might change a bit with time and w.r.t critical dimension (CD) of the pattern/design)

6. Development- MIBK:IPA (1:3): Duration- 18-22 sec (start with 15 sec)

IPA (Stopper): 30 sec

Note : Dose and development time may vary depending on the feature sizes and design

Metallization and Lift-off

1. Descum: Power- 14 watt, Gas- 10 SCCM O2, Pressure- 0.5 torr; Time - 2mins, Tool-New plasma Asher, Nano2 Lab
2. Metallization: Ti/Au: 5/55 nm; Tool- 6TEBE, 1.1 lab
3. Lift-off: Keep sample in PG Remover at 70℃ for 30-36 hrs, finally 10 sec sonication in PG remover followed by DI water rinse.

2. PMMA 950K A4 single layer recipe(+ ve tone resist)

1. Dehydration bake: 5 minute at 180 deg.
2. Spinning details:

Speed: 4000rpm, (Please check the data sheet if one needs some other resist thickness)

Pre bake: 90 sec at 180 deg.

3. Exposure:

Acc voltage: 20 KV

Dose: 150 uC/cm2- 250 uC/cm2 as per your feature size Write field: 100um

4. Development: IPA: MIBK::3:1, development time: 20s +/- 5 sec

Stopper: 2-propanol (IPA) pure, Stopping time: ~30 s

Dry: with a low-nitrogen flow

Note : Dose and development time may vary depending on the feature sizes and design

3. PMMA 950 A4 -EL9 bilayer(+ ve tone resist)

Dehydration bake: 5 minute at 180 C.

Layer1: EL9, Spin @ 3200rpm for 60 Sec

Prebake: 7min @180 C

Layer2: PMMA950K 4%, Spin@2000rpm for 45 sec

Prebake: 2min@180 C

(Please check the data sheet if one needs some other resist thickness)

Exposure:

Acc voltage: 20 KV
Dose: 150 uC/cm2- 250 uC/cm2 as per your feature size,
Write field: 100um
Developer: IPA: MIBK::3:1, developing time: 20s +/- 5 sec
Stopper: 2-propanol (IPA) pure, Stopping time: ~ 30s
Dry: with a low-nitrogen flow

Note : Dose and development time may vary depending on the feature sizes and design

4. PMMA 950 A2 -EL9 bilayer (+ ve tone resist)

Dehydration bake: 5 minutes at 180 C.
Layer1: EL9, Spin @ 3500rpm for 60 Sec
Prebake: 7min @180 C
Layer2: PMMA950K 4%, Spin@4500rpm for 45 sec
Prebake: 2min@180 C
(Please check the data sheet if one needs some other resist thickness)
Exposure:
Acc voltage: 20 KV
Dose: 130 uC/cm2- 180 uC/cm2 as per your feature size,
Write field: 100um
Developer: IPA: MIBK: 3:1, developing time: 20s +/- 5 sec
Stopper: 2-propanol (IPA) pure, Stopping time: ~ 30s
Dry: with a low-nitrogen flow

Note : Dose and development time may vary depending on the feature sizes and design

5. HSQ- High temperature (250 C) (Dow Corning XR -1541 - 6%), (-ve tone resist)

Dehydration bake: 10 minutes at 180 deg.
Spin Speed: 250 rpm, 45 sec
Prebake: 2 min at 250 deg.
Exposure:
Acc voltage: 20 KV
Dose: 400 uC- 500 uC/cm2
Write field: 100 um
Developer: 25% TMAH , Development time: 15sec-20 sec
Rinse: Flowing DI-Water, Rinsing time: 30s
Dry: with a low-nitrogen flow.

Note : Dose and development time may vary depending on the feature sizes and design

6. HSQ- Low temperature (95 C) (Dow Corning XR -1541 - 6%), (-ve tone resist)

Dehydration bake: 10 minutes at 180 deg.
Spin Speed: 2500 rpm, 45 sec
Prebake: 5 min at 95 deg C.
Exposure:
Acc voltage: 20 KV
Dose: 400 uC- 500 uC/cm2
Write field: 100 um
Developer: 25% TMAH , Development time: 20 sec-25 sec
Rinse: Flowing DI-Water, Rinsing time: 30s
Dry: with a low-nitrogen flow.
Post bake- 250 C for 15 min (for plasma etch process)

Note : Dose and development time may vary depending on the feature sizes and design

7. SU8-2000.5 (-ve tone resist)

Supplier order code: Microchem SU8-2000.5
Dehydration bake: 5 minutes at 95 deg.
Spinning: Speed: 6000rpm
Pre-bake: 65 deg- 1min - ramp to - 95 deg- 1min
Exposure-
Acc voltage: 10KV
Aperture :7.5um
Dose :4 uC/cm2
Post Bake: 65 deg- 1min - ramp to- 95 deg- 1min
Developer: SU8 developer- Developing time: 1min
Rinse: IPA- Rinsing time: 1min
Dry: with a low-nitrogen flow.
SU8 Removal
Soak in PG Remover @ 80 – 30min
Change container: Again, soak in PG remover @ 80 deg - 30min
Ultra-sonication @80deg in PG remover- 5~15 min
Time will depend on the pattern sizes

8. ZEP 520 A (+ ve tone resist)

Supplier order code: Zeon Chemicals
Resist Name: ZEP

Solvent: Anisole
Dehydration bake: 5 minutes at 180 deg.
Layer1: ZEP, Spin @ 2500 rpm for 60 Sec (~350 nm thick film)
Prebake: 3min @180 C
(Please check the data sheet if one needs some other resist thickness)
Exposure:
Acc voltage: 20KV
Dose: 45-55 uC/cm2
Aperture- depending upon feature size
Developer: ZED N50, development time: 50-60 sec
Rinse: 2-propanol (IPA) pure, Time: 30s
Post Bake – 135 C – 3 Mins (for plasma etch process)
Dry: with a low-nitrogen flow

Note : Dose and development time may vary depending on the feature sizes and design